



PSID: SIH1685

Problem Statement:

Building Offline Parallel AV Pipeline to cater multiple AVs for CII entities.

maut.

Insights.

What is need of using AV pipelines?

- No antivirus engine is perfect - a single antivirus can only detect **40-80%** of malware/viruses. [OPSWAT]
- Parallel AV pipeline:
 - a. reduces **false positives**.
 - b. renders accurate detections.

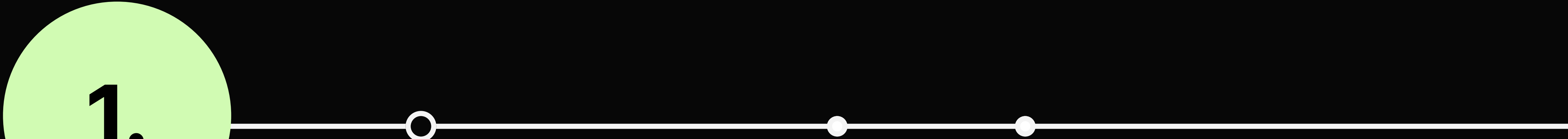
FACT: 12 Engines working simultaneously deliver a 89.6% accurate detections.

maut.

Presenting M.A.U.T.

- GUI-based cross-platform solution that performs **malware scans** using an scalable AV pipeline rendering comprehensive reports & folder organization.
- MAUT extends to provide support for:
 - **offline updates** - update history
 - scan **logs**
 - folder **monitoring** and **automatic scans**

Solution.



1.

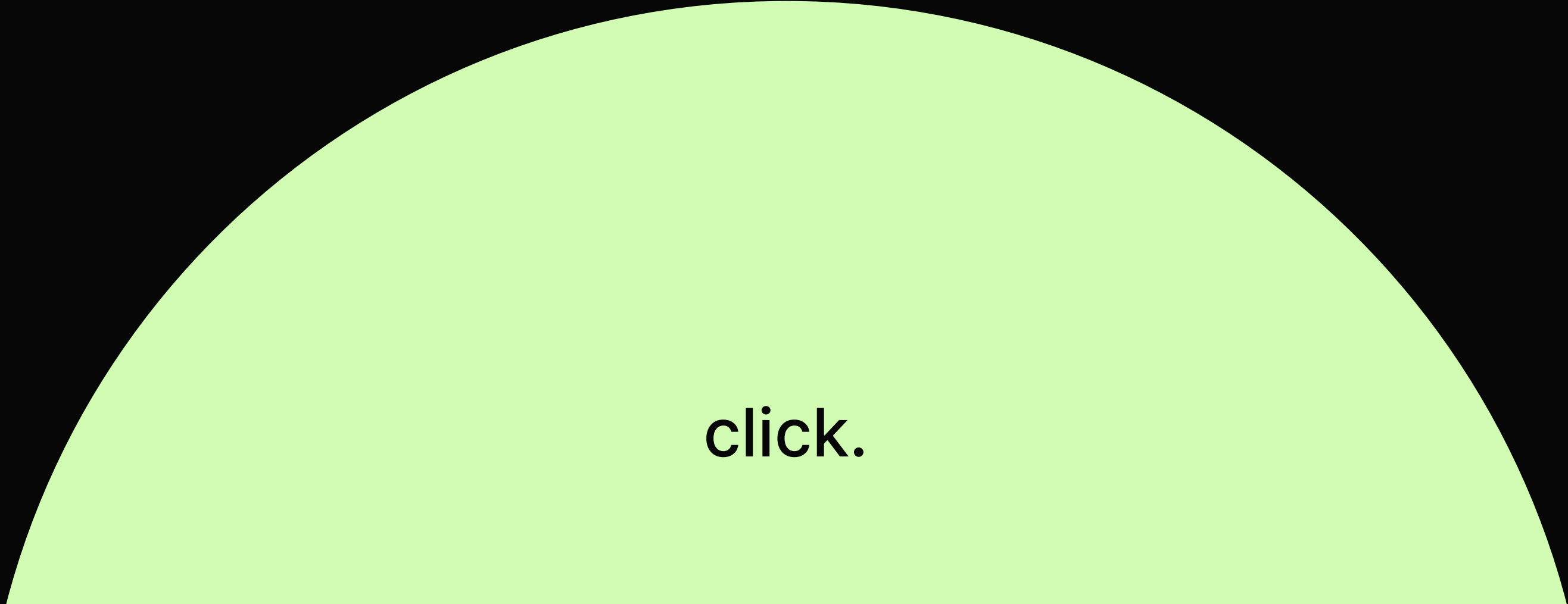
Scan.

-- select

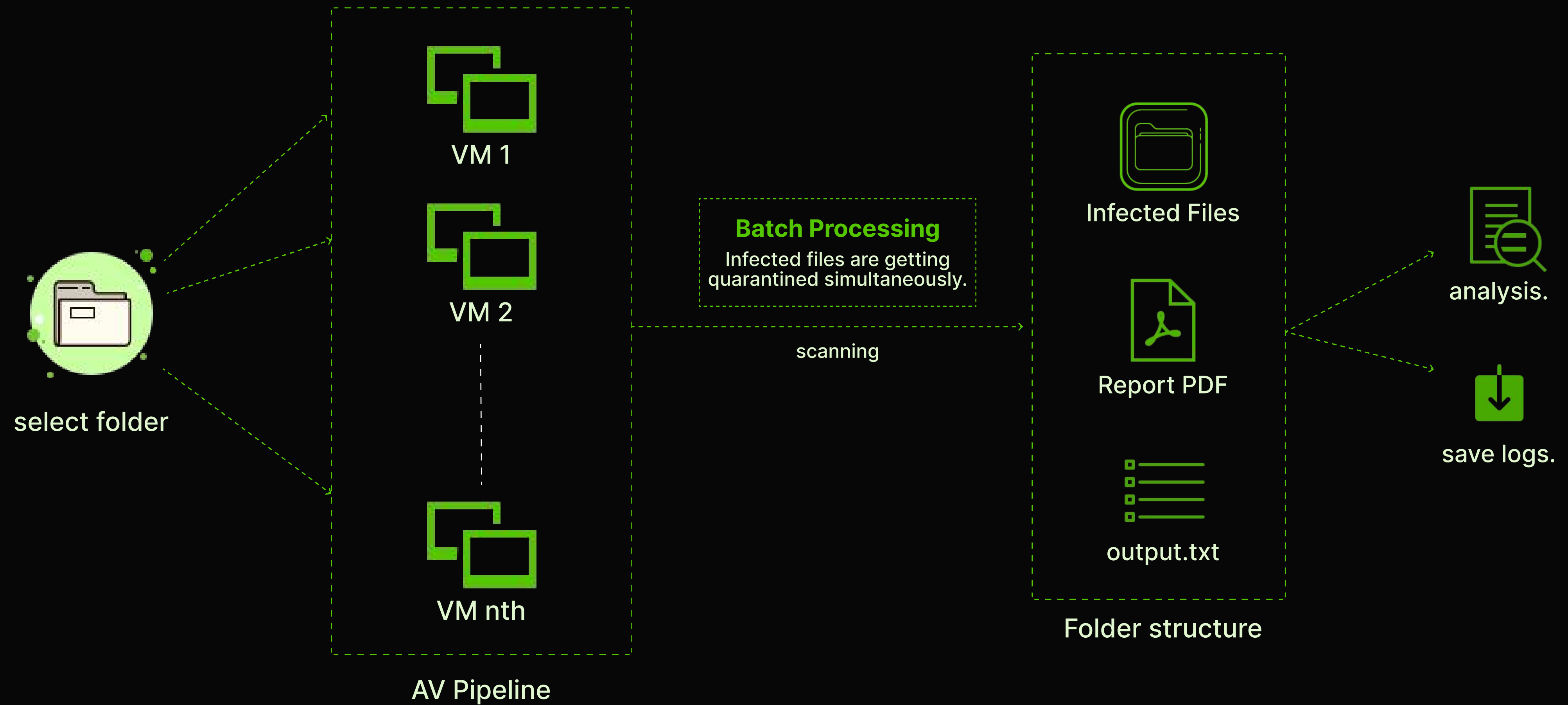
a **folder/file path**.

The files are not copied to the VM's rather
a **symlink** is created:

contains references to the folder path.



click.



Demonstration: this is the functional overview of Malware Custom Scan in M.A.U.T.

2.

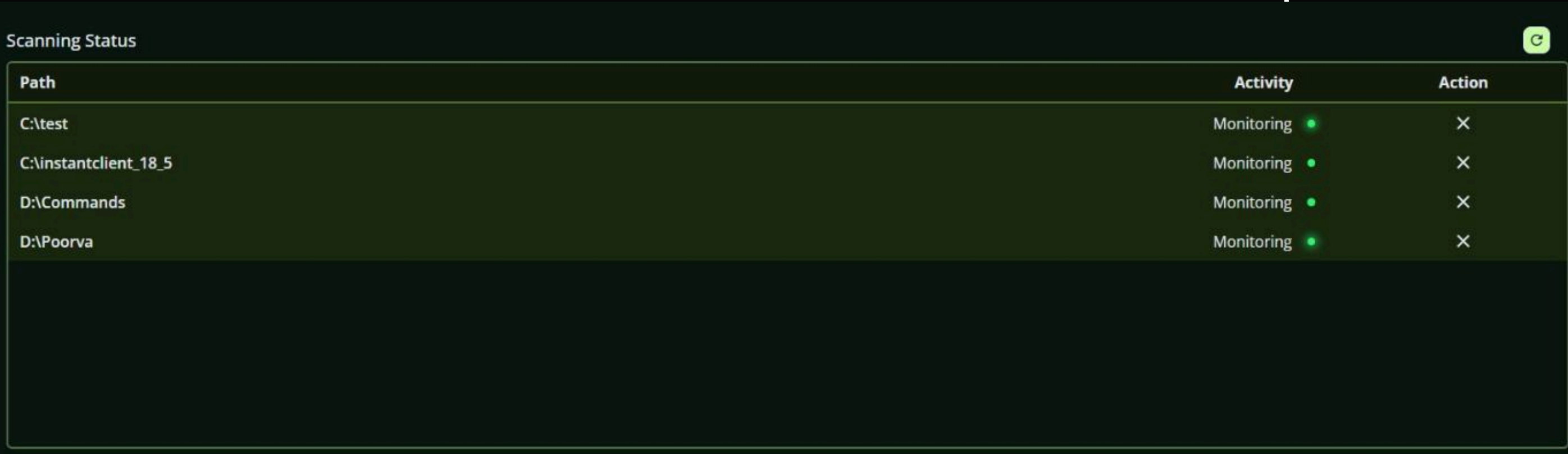
Automatic Scan.

-- folder monitoring

any folder from the host machine can be added to list and it will be monitored automatically.

```
OnChangeDetection(){  
    run CustomScan();  
}
```

click.



Path	Activity	Action
C:\test	Monitoring ●	×
C:\instantclient_18_5	Monitoring ●	×
D:\Commands	Monitoring ●	×
D:\Poorva	Monitoring ●	×

Updates.

M.A.U.T offers two kinds of updates:

a. **Offline:**

install db signatures on Internet connected systems →
transfer signatures locally →
initiate installation on AV's from **M.A.U.T.**

b. **Online:**

initiate direct online installations from the GUI

continue.

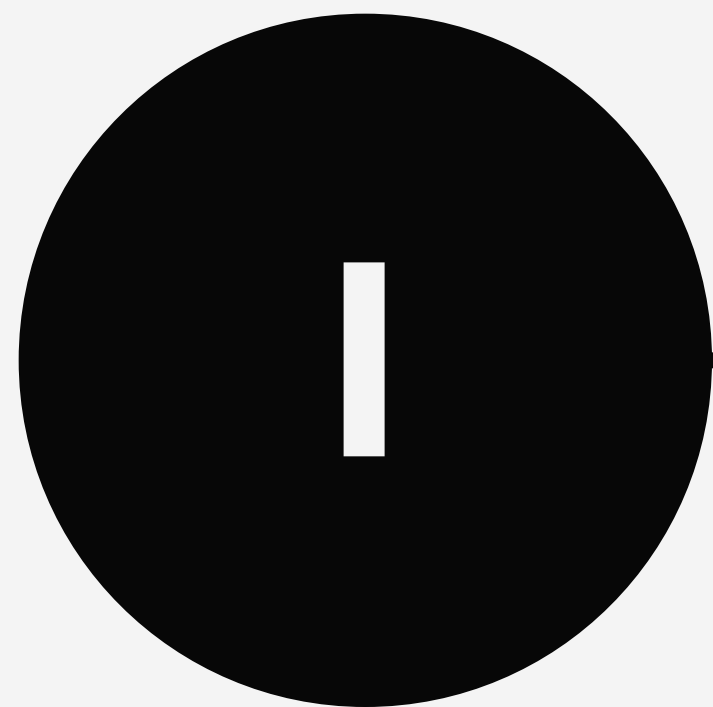
Network.

-- Network Scans.

Conduct comprehensive scans on systems connected to the host machine and keep systems in the local network safe.

- a. Connect **n** systems to the network.
- b. Resource optimization during file transfer.

network.



Results.

-- Dashboard Analysis

The dashboard contains: **combined summary**, **individual** scan results & overall **abstracted view**.

The **Analysis Page** has the following actions:

actions.



Results.



[Dashboard](#)[View Logs](#)[Updates](#)[AV Details](#)[Network](#)

Scan SummaryBACK

Time: 17:10:12**Date: 04-12-2024**

Combined Summary	Windows Defender	ESET Total Security	ClamAV
Time-taken: 38 sec Total files scanned: 5 Total Threats Found: 2	Time-taken: - Total files scanned: 5 Total Threat Files: 2	Time-taken: 1 sec (0:00:01) Total files scanned: 5 Total Threat Files: 1	Time-taken: 32.053 sec Total files scanned: 5 Total Threat Files: 1

Combined Threats⚠ Detected

Windows Defender⚠ Detected

ESET Total Security⚠ Detected

ClamAV⚠ Detected

Download Report

Logs & History.

-- track scans at every **step**.

The **View Logs** section contains a list of all scans & updates ever initiated.

1. Download **Reports**.
2. View **Analysis**.
3. Track **Update History**.



complete.

Logs & History.

M.A.U.T

Dashboard

View Logs

Updates

AV Details

Network

AntiVirus Status

ESET Internet Security

ClamAV AntiVirus

Windows Defender

Scan Logs

Scan Type	Completion Time	Action
Custom Scan [C:\\Test]	04-12-2024 / 17:00:15	Analysis Report
Custom Scan [C:\\Test]	04-12-2024 / 17:10:12	Analysis Report

Tech Stack

- Front-end **Stack-**

Tauri

TypeScript

- AntiViruses **Used-**

Windows Defender

ESET Internet Security

ClamAV

BitDefender

- Tools **used-**

VirtualBox

print pdf

guestcontrol

- Back-end **Stack-**

Rust

- Operating **System-**

Windows !!

M.A.U.T



Features

- user friendly **interface**.
- cross-platform solution.
- **folder monitoring** & **automatic** scans.
- offline **updates**.
- automatic **logs** & **history**.
- we have achieved a **scan-speed** of 3 mbps.
- easily **scalable** - integrated 4 antiviruses for testing .

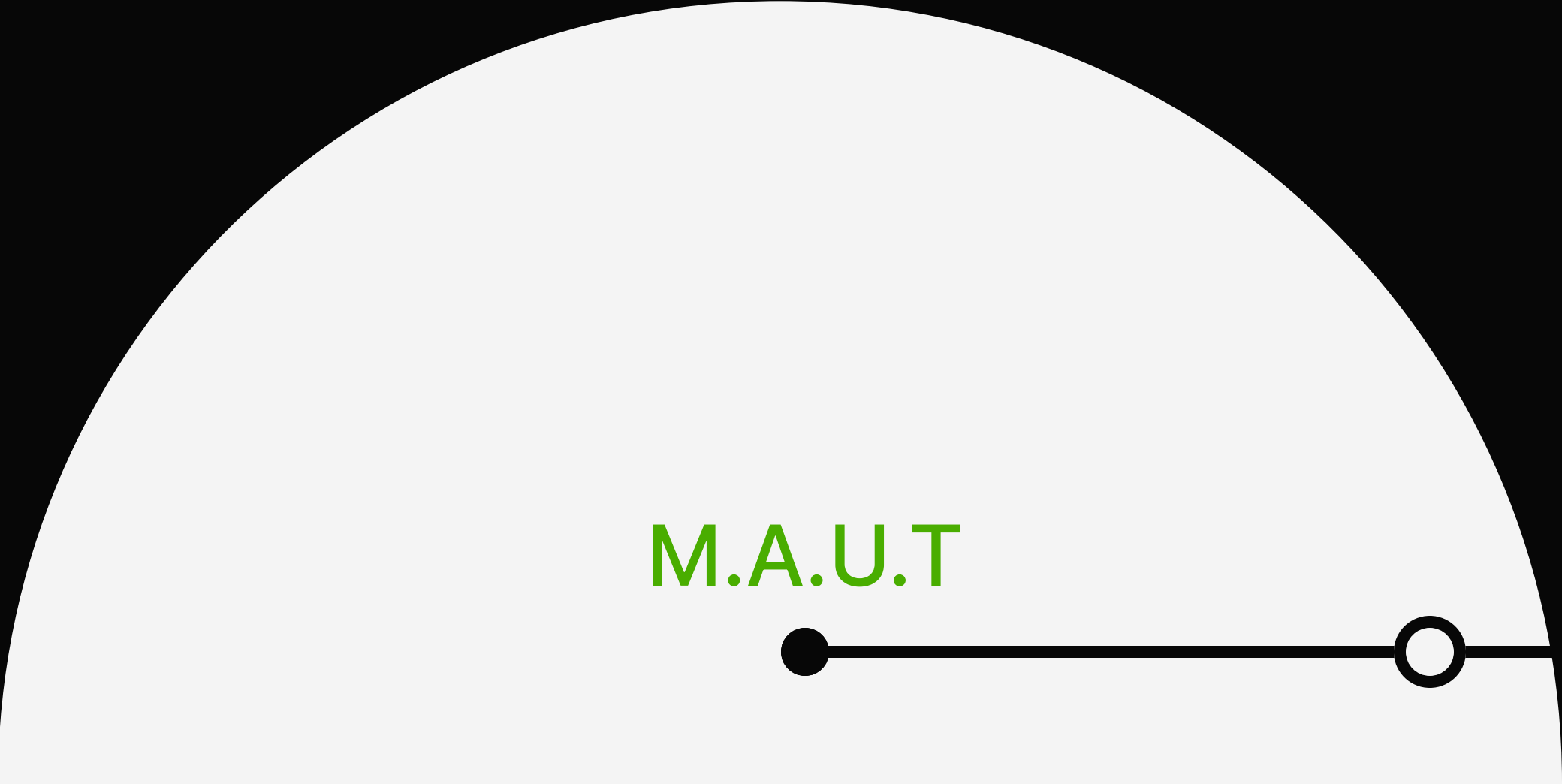
M.A.U.T



Future Scopes

- Integration of **Machine Learning** and **AI**.
- Increase in **VM** Capacity.
- Automated Scheduled **Scans**.
- Local Backup and **Restore** Functions.

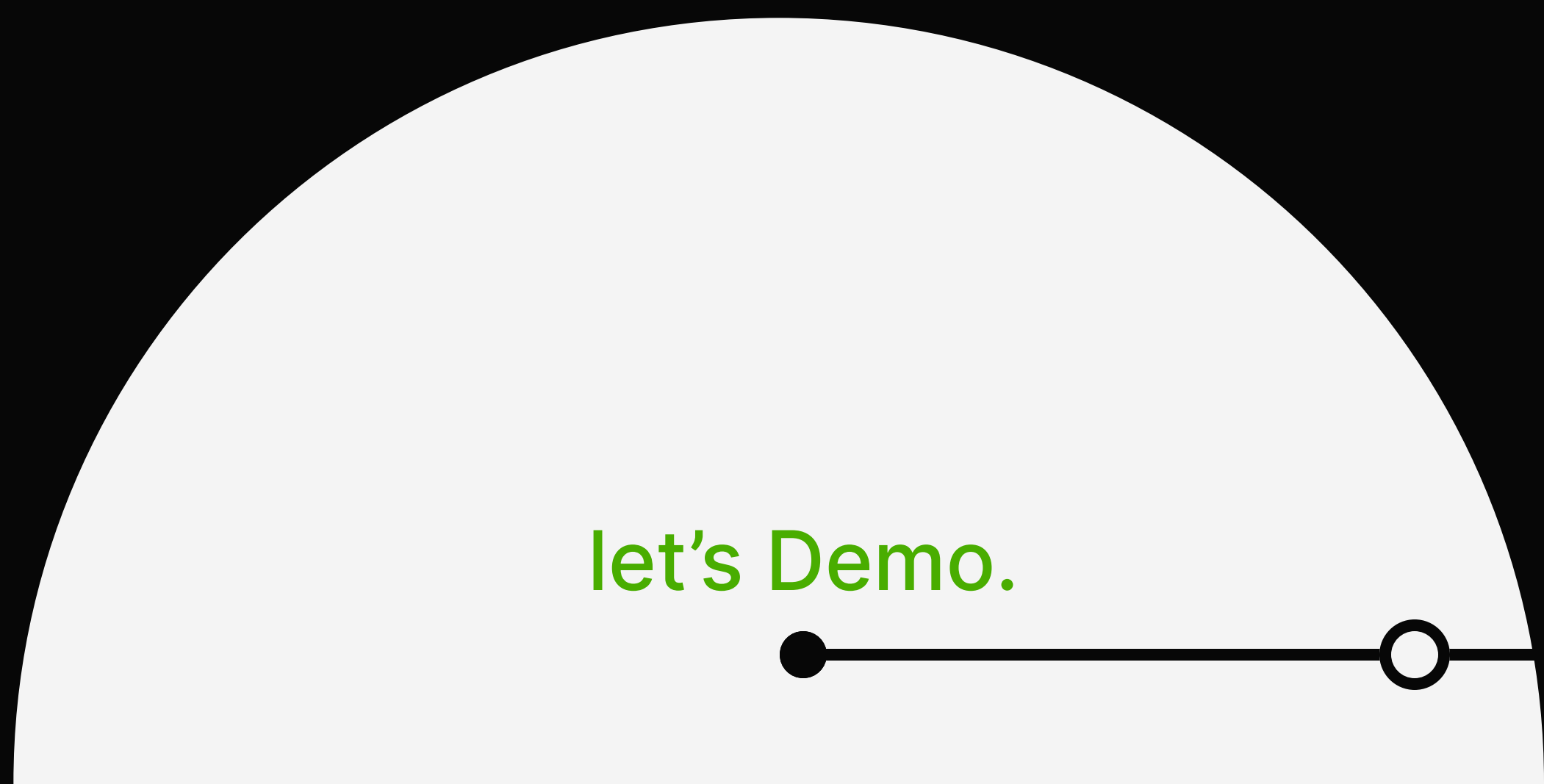
M.A.U.T





finish.

Thank you.



let's Demo.